RECOVERING THE LEVEL OF DISTRIBUTION $x^{1/2}$ FOR THE ORDERS OF $y^2 = x^3 - x$ MODULO p

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ABSTRACT. Iwaniec and Jiménez Urroz proved that for $\gg x/(\log x)^2$ primes $p \le x$, $p \equiv 1 \pmod 4$, the order of the reduction modulo p of the curve $y^2 = x^3 - x$ is eight times an integer with at most two prime factors. Xie recently pointed out that the proof of their level-of-distribution estimate is incorrect: the inclusion–exclusion that converts a rational divisibility condition $d \mid \mathrm{N}(\pi - 1)$ into congruences in $\mathbb{Z}[i]$ produces moduli of norm up to d^2, so that the Bombieri–Vinogradov theorem for $\mathbb{Q}(i)$ only gives level $x^{1/4-\varepsilon}$. We show that the estimate is nevertheless true at level $x^{1/2}(\log x)^{-B}$: the ideals of large norm are those with a large rational part, and their contribution is bounded trivially by counting lattice points. The same argument applies to the switched sequence used in the Chen-type part of the proof. Consequently the theorem of Iwaniec and Jiménez Urroz holds as stated, and Lemma 3.3 of Xie's paper holds with level $x^{1/2-\varepsilon}$; this improves her bounds for the orders over prime power fields, e.g. $|E(\mathbb{F}_{p^2})|/(4|E(\mathbb{F}_p)|)$ has at most three prime factors for $\gg x/(\log x)^2$ primes.

1. INTRODUCTION

Let E be the elliptic curve $y^2 = x^3 - x$ over $\mathbb{Q}$, which has complex multiplication by $\mathbb{Z}[i]$. For a prime $p \equiv 1 \pmod 4$ write $p = \pi\bar\pi$ with $\pi \in \mathbb{Z}[i]$ *primary*, i.e. $\pi \equiv 1 \pmod{2+2i}$; then $|E(\mathbb{F}_p)| = \mathrm{N}(\pi - 1)$, and $8 \mid \mathrm{N}(\pi - 1)$. Koblitz [10] conjectured that $|E(\mathbb{F}_p)|/8$ is prime for $\gg x/(\log x)^2$ primes $p \le x$, $p \equiv 1 \pmod 4$. Sieve methods cannot reach primes, and the strongest result one can hope for is a Chen-type theorem. Iwaniec and Jiménez Urroz [7, Theorem 1.2] state:

$$\#\Big\{p \le x :\ p \equiv 1 \pmod 4,\ \tfrac{1}{8}|E(\mathbb{F}_p)| = P_2\Big\} \gg \frac{x}{(\log x)^2}, \tag{1}$$

where P_r denotes a square-free integer with at most r prime factors. The proof follows Chen's method in $\mathbb{Z}[i]$: a weighted linear sieve applied to the sequence $\mathcal{A}(x) = \{\mathrm{N}(\pi - 1)/8\}$ ([7, §§3–4]) and a switching argument applied to the sequence $\mathcal{B}(x) = \{\mathrm{N}(1 + \omega)\}$, where ω runs over products of three Gaussian primes in prescribed ranges ([7, §5]). Both sieves need the level of distribution $x^{1/2}(\log x)^{-B}$, that is, estimates of the form

$$\sum_{d \le x^{1/2}(\log x)^{-B}} \mu^2(d)\, 3^{\omega(d)}\, |r_d(x)| \ll \frac{x}{(\log x)^A}, \tag{2}$$

where $r_d(x)$ is the error in the number of elements of the sequence divisible by d; these are (3.9) and (5.9) of [7].

In a recent paper on almost prime orders over prime power fields, Xie [12, Remark 4.1] observed that the derivation of [7, (3.9)] from the Bombieri–Vinogradov theorem for $\mathbb{Q}(i)$ is incorrect. The point is the following. A square-free d divides $\mathrm{N}(\alpha)$ if and only if, for each inert prime $q \mid d$, $q \mid \alpha$, and for each split prime $p = \pi_p\bar\pi_p$ dividing d, $\pi_p \mid \alpha$ or $\bar\pi_p \mid \alpha$. Inclusion–exclusion turns $d \mid \mathrm{N}(\alpha)$ into a signed sum of conditions $\mathfrak{a} \mid \alpha$ over the ideals $\mathfrak{a} = \prod_{q \mid d}(q) \prod_{p \mid d} \mathfrak{a}_p$

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with $\mathfrak{a}_p \in \{(\pi_p), (\bar{\pi}_p), (p)\}$, and $\mathrm{N}(\mathfrak{a})$ ranges between d and d^2. The Bombieri–Vinogradov theorem for $\mathbb{Q}(i)$ controls moduli of norm up to $x^{1/2}(\log x)^{-B}$, which only gives (2) with $d \le x^{1/4}(\log x)^{-B}$. (In [7, p. 823] the terms with $\mathfrak{a}_p = (p)$ are lost in the passage from ideals to integers; Xie's Lemma 3.3 keeps them and obtains level $x^{1/4-\varepsilon}$.) Xie concludes that (1) is not established by [7], and that the best unconditional result is P_5 [1, 12].

The purpose of this note is to show that (2) is true at level $x^{1/2}(\log x)^{-B}$ for both sequences, so that the argument of [7] is complete. The idea is simple. Write a square-free ideal $\mathfrak{a}$ as $\mathfrak{a} = (c)\mathfrak{g}$ with $c \in \mathbb{Z}_{>0}$ as large as possible (the *rational part* of $\mathfrak{a}$) and $\mathfrak{g}$ primitive. Then $d = c\,\mathrm{N}(\mathfrak{g})$ and $\mathrm{N}(\mathfrak{a}) = c\,d$. The ideals of large norm are exactly those with a large rational part, and these are rare: the expected number of elements of the sequence in a residue class modulo $\mathfrak{a}$ is of size $x/\mathrm{N}(\mathfrak{a}) = x/(cd)$, which is summable over c. For $c \le (\log x)^{B_0}$ we have $\mathrm{N}(\mathfrak{a}) \le (\log x)^{B_0} d$, which is within reach of the Bombieri–Vinogradov theorem; for $c > (\log x)^{B_0}$ the contribution is bounded trivially by counting lattice points in a disc. This is in fact the mechanism already used in [7, p. 823] to discard the inert primes, which contribute q^2 to the norm; the split primes with $\mathfrak{a}_p = (p)$ have to be treated in the same way.

We state the result in a form that applies to any sequence for which a Bombieri–Vinogradov theorem over ideals is available (Theorem 2.3), and then apply it to the two sequences of [7] (Section 3). The same statement holds over any imaginary quadratic field of class number one (Remark 2.5), which is the setting of the extension of [7] to all CM curves over $\mathbb{Q}$ in [8]. As a by-product, the level $x^{1/2-\varepsilon}$ in Lemma 3.3 of [12] improves the almost-prime bounds of [12, Theorem 1.3] for $|E(\mathbb{F}_{p^\ell})|/|E(\mathbb{F}_p)|$ (Corollary 3.3). Section 4 records a few remarks on the remaining parts of [7], in particular on the hypothesis of their Lemma 5.2. The code and data behind the numerical statements are described at the end of the paper (GitHub and Zenodo, doi:10.5281/zenodo.23428952).

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2. The main lemma

Throughout, $\mathrm{N}(\alpha) = |\alpha|^2$ is the norm on $\mathbb{Z}[i]$, ideals are denoted by fraktur letters, $\Phi(\mathfrak{m}) = |(\mathbb{Z}[i]/\mathfrak{m})^\times|$, and $L = \log x$. Every ideal of $\mathbb{Z}[i]$ is principal.

2.1. Lattice points in a residue class.

Lemma 2.1. *Let $\mathfrak{m} = (\gamma) \neq 0$ be an ideal of $\mathbb{Z}[i]$, $\alpha \in \mathbb{Z}[i]$ and $r > 0$. Then*

$$\#\{w \in \alpha + \mathfrak{m} : \ |w| \le r\} \le \Big(\frac{2r}{\mathrm{N}(\mathfrak{m})^{1/2}} + 1\Big)^2 \ll \frac{r^2}{\mathrm{N}(\mathfrak{m})} + 1.$$

Proof. The set $\alpha + \mathfrak{m} = \alpha + \gamma\mathbb{Z}[i]$ is a translate of the lattice $\gamma\mathbb{Z}[i] \subset \mathbb{C}$, whose nonzero vectors have length at least $\lambda = |\gamma| = \mathrm{N}(\mathfrak{m})^{1/2}$. The open discs of radius $\lambda/2$ centred at the points of $\alpha + \mathfrak{m}$ are pairwise disjoint, and those centred at points of modulus $\le r$ are contained in the disc of radius $r + \lambda/2$. Comparing areas gives the bound $(r + \lambda/2)^2/(\lambda/2)^2$. □

2.2. The rational part of a square-free ideal.

For an ideal $\mathfrak{a} \neq 0$ let $\mathrm{d}(\mathfrak{a})$ be the least positive rational integer in $\mathfrak{a}$, and let $\mathrm{c}(\mathfrak{a})$ be the largest positive rational integer c with $(c) \supseteq \mathfrak{a}$. Then $\mathfrak{a} = (\mathrm{c}(\mathfrak{a}))\,\mathfrak{g}$ where $\mathfrak{g}$ is *primitive*, i.e. not divisible by any rational integer > 1.

Lemma 2.2. *Let $\mathfrak{a}$ be a square-free ideal, $c = \mathrm{c}(\mathfrak{a})$ and $\mathfrak{a} = (c)\mathfrak{g}$. Then c is odd and square-free, $\mathrm{d}(\mathfrak{a}) = c\,\mathrm{N}(\mathfrak{g})$, and*

$$\mathrm{N}(\mathfrak{a}) = \mathrm{c}(\mathfrak{a})\,\mathrm{d}(\mathfrak{a}).$$

Moreover, for a square-free integer $d \geq 1$ the square-free ideals $\mathfrak{a}$ with $\mathrm{d}(\mathfrak{a}) = d$ are exactly the ideals

$$\mathfrak{a} = (1+i)^e \prod_{q \mid d} (q) \prod_{p \mid d} \mathfrak{a}_p, \qquad e = 1 \text{ if } 2 \mid d,\ e = 0 \text{ otherwise}, \qquad \mathfrak{a}_p \in \{(\pi_p), (\bar{\pi}_p), (p)\},$$

where q runs over the primes $\equiv 3 \pmod 4$ and $p = \pi_p \bar{\pi}_p$ over the primes $\equiv 1 \pmod 4$ dividing d. In particular their number is $3^{\omega_1(d)} \leq 3^{\omega(d)}$, where $\omega_1(d)$ is the number of prime factors $p \equiv 1 \pmod 4$ of d, and $\mathrm{c}(\mathfrak{a}) = \prod_{q \mid d} q \prod_{\mathfrak{a}_p = (p)} p$ divides d.

Proof. Since $\mathfrak{a}$ is square-free and $(2) = (1+i)^2$, no even integer c satisfies $(c) \supseteq \mathfrak{a}$, and c is square-free. A rational integer n lies in $\mathfrak{g}$ if and only if every prime ideal factor $\mathfrak{p}$ of $\mathfrak{g}$ divides n; for $\mathfrak{p} = (\pi_p)$ or $(\bar{\pi}_p)$ this means $p \mid n$ (as $n = \bar{n}$), for $\mathfrak{p} = (q)$ it means $q \mid n$, and for $\mathfrak{p} = (1+i)$ it means $2 \mid n$. Since $\mathfrak{g}$ is primitive and square-free, no two of its prime ideal factors lie above the same rational prime, so $n \in \mathfrak{g}$ if and only if $\mathrm{N}(\mathfrak{g}) \mid n$, i.e. $\mathrm{d}(\mathfrak{g}) = \mathrm{N}(\mathfrak{g})$. Now $n \in (c)\mathfrak{g}$ if and only if $c \mid n$ and $n/c \in \mathfrak{g}$, so $\mathrm{d}(\mathfrak{a}) = c\,\mathrm{N}(\mathfrak{g})$, and $\mathrm{N}(\mathfrak{a}) = c^2 \mathrm{N}(\mathfrak{g}) = c\,\mathrm{d}(\mathfrak{a})$. The description of the ideals with $\mathrm{d}(\mathfrak{a}) = d$ follows from $\mathrm{d}(\mathfrak{a}) = \prod_{\mathfrak{p} \mid \mathfrak{a}} \mathrm{d}(\mathfrak{p})$ for square-free $\mathfrak{a}$ (the primes above different rational primes being coprime), with $\mathrm{d}((\pi_p)) = \mathrm{d}((\bar{\pi}_p)) = \mathrm{d}((p)) = p$, $\mathrm{d}((q)) = q$, $\mathrm{d}((1+i)) = 2$, and (2) being excluded as not square-free. □

2.3. **From norm to least integer.**

Theorem 2.3. *Let $x \geq 3$ and let $R(\mathfrak{m}) \geq 0$ be a function on the nonzero ideals of $\mathbb{Z}[i]$ such that*

(i) *for every $A > 0$ there is $B = B(A) > 0$ with*

$$\sum_{\mathrm{N}(\mathfrak{m}) \leq x^{1/2} L^{-B}} R(\mathfrak{m}) \ll_A \frac{x}{L^A};$$

(ii) $R(\mathfrak{m}) \ll xL/\mathrm{N}(\mathfrak{m}) + 1$ *for every $\mathfrak{m}$ with $\mathrm{N}(\mathfrak{m}) \leq x$.*

Let $\mathfrak{b}$ be a fixed nonzero ideal and $k \geq 0$. Then for every $A > 0$ there is $B' = B'(A, k, \mathfrak{b})$ such that

$$\sum_{\substack{\mathfrak{a} \text{ square-free} \\ \mathrm{d}(\mathfrak{a}) \leq x^{1/2} L^{-B'}}} \tau(\mathrm{d}(\mathfrak{a}))^k\, R(\mathfrak{b}\mathfrak{a}) \ll_{A,k,\mathfrak{b}} \frac{x}{L^A}.$$

Proof. Put $D = x^{1/2} L^{-B'}$ and $C_0 = L^{B_0}$, with B_0, B' to be chosen. Note that every ideal in the sum satisfies $\mathrm{N}(\mathfrak{b}\mathfrak{a}) = \mathrm{N}(\mathfrak{b})\mathrm{c}(\mathfrak{a})\mathrm{d}(\mathfrak{a}) \leq \mathrm{N}(\mathfrak{b}) D^2 \leq x$ for x large, since $\mathrm{c}(\mathfrak{a}) \leq \mathrm{d}(\mathfrak{a})$; so hypothesis (ii) is available throughout. Split the sum according to whether $\mathrm{c}(\mathfrak{a}) \leq C_0$ or $\mathrm{c}(\mathfrak{a}) > C_0$.

Ideals with small rational part. If $\mathrm{c}(\mathfrak{a}) \leq C_0$ and $\mathrm{d}(\mathfrak{a}) \leq D$, then by Lemma 2.2

$$\mathrm{N}(\mathfrak{b}\mathfrak{a}) = \mathrm{N}(\mathfrak{b})\,\mathrm{c}(\mathfrak{a})\,\mathrm{d}(\mathfrak{a}) \leq \mathrm{N}(\mathfrak{b})\, C_0 D \leq x^{1/2} L^{-B}$$

provided $B' \geq B + B_0 + 1$ and x is large in terms of $\mathfrak{b}$, where $B = B(A_1)$ with A_1 to be chosen. Since $\mathfrak{a} \mapsto \mathfrak{b}\mathfrak{a}$ is injective and $R \geq 0$, hypothesis (i) gives $\sum R(\mathfrak{b}\mathfrak{a}) \ll xL^{-A_1}$ over this family. To insert the weight $\tau(\mathrm{d}(\mathfrak{a}))^k$ we use Cauchy's inequality together with the trivial bound (ii):

$$\sum \tau(\mathrm{d}(\mathfrak{a}))^k R(\mathfrak{b}\mathfrak{a}) \leq \Big(\sum \tau(\mathrm{d}(\mathfrak{a}))^{2k} R(\mathfrak{b}\mathfrak{a}) \Big)^{1/2} \Big(\sum R(\mathfrak{b}\mathfrak{a}) \Big)^{1/2},$$

and, grouping the ideals according to $d = \mathrm{d}(\mathfrak{a})$ (at most $3^{\omega(d)}$ ideals for each d) and using $\mathrm{N}(\mathfrak{b}\mathfrak{a}) \geq \mathrm{d}(\mathfrak{a})$,

$$\sum \tau(\mathrm{d}(\mathfrak{a}))^{2k} R(\mathfrak{b}\mathfrak{a}) \ll \sum_{d \leq D} \tau(d)^{2k} 3^{\omega(d)} \Big(\frac{xL}{d} + 1 \Big) \ll xL^{c_k}$$

for some c_k depending only on k. Hence the ideals with small rational part contribute $\ll xL^{(c_k-A_1)/2}$, which is $\ll xL^{-A}$ for $A_1 = 2A + c_k$.

Ideals with large rational part. For $\mathrm{c}(\mathfrak{a}) > C_0$ we use (ii) and $\mathrm{N}(\mathfrak{b}\mathfrak{a}) \geq \mathrm{N}(\mathfrak{a}) = \mathrm{c}(\mathfrak{a})\mathrm{d}(\mathfrak{a}) > C_0\,\mathrm{d}(\mathfrak{a})$, so that $R(\mathfrak{b}\mathfrak{a}) \ll xL/(C_0\mathrm{d}(\mathfrak{a})) + 1$. Grouping again according to $d = \mathrm{d}(\mathfrak{a})$ (at most $3^{\omega(d)}$ ideals for each d), the contribution is

$$\ll \sum_{d \leq D} \tau(d)^k\, 3^{\omega(d)} \Big(\frac{xL}{C_0 d} + 1 \Big) \ll \frac{xL^{c'_k}}{C_0} + DL^{c'_k},$$

which is $\ll xL^{-A}$ once $B_0 = A + c'_k$. This proves the theorem with $B' = B(2A + c_k) + A + c'_k + 1$. □

Remark 2.4. The hypothesis (ii) is a trivial bound, satisfied by every R that arises as the discrepancy of a set of Gaussian integers of norm $\leq x$ in residue classes, by Lemma 2.1. The proof uses (ii) only inside sums carrying divisor-function weights, so (ii) may be weakened to $R(\mathfrak{m}) \ll \tau(\mathrm{N}(\mathfrak{m}))^c\,(xL/\mathrm{N}(\mathfrak{m}) + 1)$ for a fixed c, at the cost of changing c_k, c'_k: for $\mathfrak{m} = \mathfrak{b}\mathfrak{a}$ with $\mathrm{d}(\mathfrak{a}) = d$ one has $\mathrm{N}(\mathfrak{m}) \mid \mathrm{N}(\mathfrak{b})d^2$, hence $\tau(\mathrm{N}(\mathfrak{m})) \leq \tau(\mathrm{N}(\mathfrak{b}))\tau(d)^2$. This covers sets defined by a condition $\mathfrak{a} \mid \Phi(\pi)$ with Φ a polynomial, where the number of admissible residue classes modulo $\mathfrak{a}$ is $\ll \deg(\Phi)^{\omega(\mathfrak{a})}$. Hypothesis (i) is a Bombieri–Vinogradov theorem over ideals of norm up to $x^{1/2}L^{-B}$. The conclusion is a Bombieri–Vinogradov theorem over ideals of *least integer* up to $x^{1/2}L^{-B'}$, which is what a rational sieve applied to $\mathrm{N}(\alpha)$ needs. Nothing is lost: in the main term of the sieve the ideals with large rational part contribute $\ll xL^c/C_0$ as well, so they are negligible on both sides.

Remark 2.5. Lemmas 2.1 and 2.2 and Theorem 2.3 hold for the ring of integers $\mathcal{O}_K$ of any imaginary quadratic field K of class number one, with $\mathbb{Z}[i]$ replaced by $\mathcal{O}_K \subset \mathbb{C}$ and with one change of wording: every ideal (γ) is a lattice with minimal distance $|\gamma| = \mathrm{N}(\gamma)^{1/2}$, and the decomposition $\mathfrak{a} = (c)\mathfrak{g}$ with $\mathrm{N}(\mathfrak{a}) = c\,\mathrm{d}(\mathfrak{a})$ holds for square-free $\mathfrak{a}$, where now c is square-free and coprime to the discriminant of K (a ramified prime $(p) = \mathfrak{p}^2$ occurs at most once in $\mathfrak{a}$ and contributes its norm p to $\mathrm{d}(\mathfrak{g}) = \mathrm{N}(\mathfrak{g})$; in $\mathbb{Z}[i]$ the only ramified prime is 2, whence "c is odd" in Lemma 2.2). This is the setting of the extension of [7] to all CM curves over $\mathbb{Q}$ in [8], which follows the argument of [7] and is therefore affected by the same gap; Theorem 2.3 supplies the missing estimate there as well, but we have not checked the remaining details of [8].

3. Application to the two sequences of Iwaniec and Jiménez Urroz

We keep the notation of [7]: $z = x^{1/8}$, $y = x^{1/3}$, and π_1, π_2, π_3 denote primary Gaussian primes above split rational primes, $\mathrm{N}(\pi_j) = p_j \equiv 1 \pmod 4$. A primary prime π is either of this kind or $\pi = -q$ with $q \equiv 3 \pmod 4$ inert; there are $O(x^{1/2})$ of the latter with $\mathrm{N}(\pi) \leq x$, and we include them in the sequence $\mathcal{A}(x)$ below (see the comment after Corollary 3.1).

3.1. **Inclusion–exclusion.** For $\alpha \in \mathbb{Z}[i]$ and a prime ℓ we have: $2 \mid \mathrm{N}(\alpha)$ iff $(1+i) \mid \alpha$; for $q \equiv 3 \pmod 4$, $q \mid \mathrm{N}(\alpha)$ iff $q \mid \alpha$; for $p = \pi_p\bar{\pi}_p \equiv 1 \pmod 4$, $p \mid \mathrm{N}(\alpha)$ iff $\pi_p \mid \alpha$ or $\bar{\pi}_p \mid \alpha$, so that

$$\mathbf{1}_{p \mid \mathrm{N}(\alpha)} = \mathbf{1}_{\pi_p \mid \alpha} + \mathbf{1}_{\bar{\pi}_p \mid \alpha} - \mathbf{1}_{p \mid \alpha}.$$

Multiplying over the prime factors of a square-free d and using Lemma 2.2 we obtain

$$\mathbf{1}_{d \mid \mathrm{N}(\alpha)} = \sum_{\substack{\mathfrak{a} \text{ square-free} \\ \mathrm{d}(\mathfrak{a}) = d}} (-1)^{r(\mathfrak{a})}\, \mathbf{1}_{\mathfrak{a} \mid \alpha}, \qquad r(\mathfrak{a}) = \#\{p \mid d :\ \mathfrak{a}_p = (p)\}. \tag{3}$$

This is [12, Lemma 3.1 and (3.12)]. Define the multiplicative function $\hat{g}$ on square-free ideals by $\hat{g}(\mathfrak{a}) = 4/\Phi((2+2i)\mathfrak{a})$, so that $\hat{g}((1+i)) = 1/2$, $\hat{g}((\pi_p)) = \hat{g}((\bar{\pi}_p)) = 1/(p-1)$, $\hat{g}((q)) = 1/(q^2-1)$, and

(4)
$$g(d) = \sum_{\mathrm{d}(\mathfrak{a})=d} (-1)^{r(\mathfrak{a})}\hat{g}(\mathfrak{a}) = \prod_{\ell \mid d} g(\ell), \qquad g(2) = \tfrac{1}{2}, \quad g(q) = \frac{1}{q^2-1}, \quad g(p) = \frac{2}{p-1} - \frac{1}{(p-1)^2},$$

which is the density function of [7, (3.7)].

3.2. **The sequence $\mathcal{A}(x)$.** Let $\Pi^*(x)$ be the number of prime elements $\pi \in \mathbb{Z}[i]$ (all associates counted) with $\mathrm{N}(\pi) \le x$, and for an ideal $\mathfrak{q}$ and $\alpha \in \mathbb{Z}[i]$ let $\Pi(x; \mathfrak{q}, \alpha)$ be the number of those with $\pi \equiv \alpha \pmod{\mathfrak{q}}$. The Bombieri–Vinogradov theorem for $\mathbb{Q}(i)$ (Huxley [6, Theorem 1], stated there for reduced narrow ideal classes of a number field; the form below for residue classes in $\mathbb{Z}[i]$ is [12, Proposition 2.2] and [7, Proposition 3.2]; see also Johnson [9, Corollary] and Hinz [4]) states that for every A there is B with

$$\sum_{\mathrm{N}(\mathfrak{q}) \le x^{1/2} L^{-B}} \max_{(\alpha,\mathfrak{q})=1} \Bigl| \Pi(x; \mathfrak{q}, \alpha) - \frac{\Pi^*(x)}{\Phi(\mathfrak{q})} \Bigr| \ll \frac{x}{L^A}. \tag{5}$$

(For $\mathrm{N}(\mathfrak{q}) \le 4$ the terms are trivially $O(x^{1/2})$; for $\mathrm{N}(\mathfrak{q}) > 4$ the four units are distinct modulo $\mathfrak{q}$, so counting prime elements in a class is the same as counting prime ideals with a generator in that class, as in [12, Proposition 2.2].) Let $X = \Pi^*(x)/4$; this is the number of prime ideals of norm $\le x$, and $X = \Pi(x) + O(x^{1/2})$ in the notation of [7], where $\Pi(x)$ counts the primary primes above split $p \le x$. For square-free d let

$$A_d(x) = \#\{\pi \text{ primary} : \ \mathrm{N}(\pi) \le x, \ d \mid \mathrm{N}(\pi - 1)/8\}.$$

Since π is primary, $\alpha = (\pi - 1)/(2 + 2i) \in \mathbb{Z}[i]$ and $\mathrm{N}(\alpha) = \mathrm{N}(\pi - 1)/8$; moreover $\mathfrak{a} \mid \alpha$ if and only if $\pi \equiv 1 \pmod{(2+2i)\mathfrak{a}}$, and the condition $\pi \equiv 1 \pmod{2+2i}$ is then automatic. Hence by (3)

$$A_d(x) = \sum_{\mathrm{d}(\mathfrak{a})=d} (-1)^{r(\mathfrak{a})}\, \Pi\bigl(x; (2+2i)\mathfrak{a}, 1\bigr) = X g(d) + r_d(x), \qquad |r_d(x)| \le \sum_{\mathrm{d}(\mathfrak{a})=d} R_{\mathcal{A}}((2+2i)\mathfrak{a}),$$

where $R_{\mathcal{A}}(\mathfrak{q}) = \max_{(\alpha,\mathfrak{q})=1} |\Pi(x; \mathfrak{q}, \alpha) - \Pi^*(x)/\Phi(\mathfrak{q})|$. Hypothesis (i) of Theorem 2.3 for $R_{\mathcal{A}}$ is (5); hypothesis (ii) follows from Lemma 2.1, which gives $\Pi(x; \mathfrak{q}, \alpha) \ll x/\mathrm{N}(\mathfrak{q}) + 1$, and from $\Phi(\mathfrak{q}) \gg \mathrm{N}(\mathfrak{q})/\log\log \mathrm{N}(\mathfrak{q})$. Theorem 2.3 with $\mathfrak{b} = (2+2i)$ and $k = 2$ (note $3^{\omega(d)} \le \tau(d)^2$) gives:

Corollary 3.1. *For every $A > 0$ there is $B' > 0$ such that*

$$\sum_{d \le x^{1/2} L^{-B'}} \mu^2(d)\, 3^{\omega(d)}\, |r_d(x)| \ll \frac{x}{L^A}.$$

This is [7, (3.9)], with the weight $3^{\omega(d)}$ required by the Jurkat–Richert theorem in the form of [3, Theorem 8.4]. (In [7] the sequence $\mathcal{A}(x)$ is indexed by the primary primes above split p only; our $A_d(x)$ also counts the $O(x^{1/2})$ primary primes $-q$, $q \equiv 3 \pmod 4$, $q \le x^{1/2}$, for which the identity above is exact. Removing them changes the sifted sum by at most $O(x^{1/2})$, which is negligible against $x/(\log x)^2$.) It is also Lemma 3.3 of [12] with the condition $(d_1 d_2)^2 \le x^{1/2} L^{-B}$ replaced by $d_1 d_2 \le x^{1/2} L^{-B'}$ (the vector-sieve remainder of [12, Proposition 2.8] involves the same family of moduli).

3.3. **The switched sequence $\mathcal{B}(x)$.** Let

$$\Omega(x)=\big\{\omega=(1+i)^3\varepsilon\,\pi_1\pi_2\pi_3:\ \varepsilon\in\{\pm1,\pm i\},\ \mathrm{N}(\omega)\le x,\ z\le \mathrm{N}(\pi_3)<y\le \mathrm{N}(\pi_2)<\mathrm{N}(\pi_1)\big\}.$$

Since the norms of π_1,π_2,π_3 are distinct, no conjugate pair occurs among them, and by unique factorisation the map $\omega\mapsto(\varepsilon,\pi_1,\pi_2,\pi_3)$ is injective; thus $\Omega(x)$ is a set of $\ll x$ Gaussian integers of norm $\le x$, each counted once. All elements $1+\omega$ are coprime to $1+i$, so $\mathrm{N}(1+\omega)$ is odd. For an odd ideal $\mathfrak{m}$ and $(\alpha,\mathfrak{m})=1$ put

$$E(x;\alpha,\mathfrak{m})=\#\{\omega\in\Omega(x):\ \omega\equiv\alpha\ (\mathrm{mod}\ \mathfrak{m})\}-\frac{1}{\Phi(\mathfrak{m})}\#\{\omega\in\Omega(x):\ (\omega,\mathfrak{m})=1\},$$

and $R_{\mathcal{B}}(\mathfrak{m})=\max_{(\alpha,\mathfrak{m})=1}|E(x;\alpha,\mathfrak{m})|$ for odd $\mathfrak{m}$, $R_{\mathcal{B}}(\mathfrak{m})=0$ for even $\mathfrak{m}$ (both counts vanish then). Proposition 5.1 of [7] is the statement that $R_{\mathcal{B}}$ satisfies hypothesis (i) of Theorem 2.3; see Section 4 for its proof. Hypothesis (ii) is again Lemma 2.1, since $|E(x;\alpha,\mathfrak{m})|\le\#\{w\in\alpha+\mathfrak{m}:\mathrm{N}(w)\le x\}+|\Omega(x)|/\Phi(\mathfrak{m})$.

For odd square-free d let $B_d(x)=\#\{\omega\in\Omega(x):d\mid\mathrm{N}(1+\omega)\}$. By (3) with $\alpha=1+\omega$,

$$B_d(x)=\sum_{\mathrm{d}(\mathfrak{a})=d}(-1)^{r(\mathfrak{a})}\#\{\omega\in\Omega(x):\ \omega\equiv-1\ (\mathrm{mod}\ \mathfrak{a})\}=|\Omega(x)|\,g(d)+r_d^{\mathcal{B}}(x),$$

where now, since $\Phi(\mathfrak{a})^{-1}=\hat{g}(\mathfrak{a})$ for odd $\mathfrak{a}$,

$$r_d^{\mathcal{B}}(x)=\sum_{\mathrm{d}(\mathfrak{a})=d}(-1)^{r(\mathfrak{a})}\Big(E(x;-1,\mathfrak{a})-\frac{1}{\Phi(\mathfrak{a})}\#\{\omega\in\Omega(x):\ (\omega,\mathfrak{a})\ne1\}\Big).$$

The first part is handled by Theorem 2.3 with $\mathfrak{b}=(1)$. For the second part note that $(\omega,\mathfrak{a})\ne1$ forces some $\pi_j\mid\mathfrak{a}$, with $\mathrm{N}(\pi_j)\ge z$, and for a fixed prime π the number of $\omega\in\Omega(x)$ divisible by π is at most $4\#\{\beta:\mathrm{N}(\beta)\le x/(8\mathrm{N}(\pi))\}\ll x/\mathrm{N}(\pi)$. Hence, writing $\mathfrak{a}=(\pi)\mathfrak{a}'$,

$$\sum_{\mathrm{d}(\mathfrak{a})\le D}3^{\omega(\mathrm{d}(\mathfrak{a}))}\frac{\#\{\omega:(\omega,\mathfrak{a})\ne1\}}{\Phi(\mathfrak{a})}\ll x\sum_{\mathrm{N}(\pi)\ge z}\frac{1}{\mathrm{N}(\pi)\Phi((\pi))}\sum_{\mathrm{N}(\mathfrak{a}')\le D^2}\frac{3^{\omega(\mathrm{d}(\mathfrak{a}'))}}{\Phi(\mathfrak{a}')}\ll\frac{xL^c}{z}.$$

Corollary 3.2. *For every $A>0$ there is $B'>0$ such that*

$$\sum_{\substack{d\le x^{1/2}L^{-B'}\\ d\ \text{odd}}}\mu^2(d)\,3^{\omega(d)}\,|r_d^{\mathcal{B}}(x)|\ll\frac{x}{L^A}.$$

This is [7, (5.9)], whose derivation from Proposition 5.1 there is said to be "the same as the one done to go from (3.8) to (3.9)" and therefore had the same gap.

3.4. **Conclusion.** Corollaries 3.1 and 3.2 are precisely the inputs [7, (3.9) and (5.9)] needed for the three sieve estimates [7, (3.13), (4.3), (5.10)], namely

$$W_1(x)\ge\Big(\tfrac12e^\gamma\log3-\varepsilon\Big)\Pi(x)V(z),\qquad W_2(x)\le\Big(\tfrac12e^\gamma\log6+\varepsilon\Big)\Pi(x)V(z),$$

$$W_3(x)\le\Big(\tfrac12e^\gamma c+\varepsilon\Big)\Pi(x)V(z),\qquad c=\int_{1/8}^{1/3}\int_{1/3}^{(1-u)/2}\frac{\mathrm{d}u\,\mathrm{d}v}{uv(1-u-v)}=0.36308372\ldots,$$

and therefore

$$\begin{aligned}W_1-\tfrac12W_2-\tfrac12W_3&\ge\tfrac12e^\gamma\big(\log3-\tfrac12\log6-\tfrac12c-3\varepsilon\big)\Pi(x)V(z)\\&=\tfrac12e^\gamma(0.02119\ldots-3\varepsilon)\,\Pi(x)V(z)>0.\end{aligned}$$

Hence (1) holds: *Theorem 1.2 of* [7] *is correct as stated.*

3.5. **Orders over prime power fields.** Theorem 2.3 also raises the level of distribution in the prime-power results of [12]. For an odd prime ℓ, $|E(\mathbb{F}_{p^\ell})|/|E(\mathbb{F}_p)| = \mathrm{N}(\Phi_\ell(\pi))$ with Φ_ℓ the ℓ-th cyclotomic polynomial, and for $\ell = 2$, $|E(\mathbb{F}_{p^2})|/(4|E(\mathbb{F}_p)|) = \mathrm{N}(\pi+1)/4$. In [12, §4] these sequences are sifted with a weighted linear sieve whose remainder is controlled by [12, (4.6)], an estimate of the form (2) at level $x^{1/4-\varepsilon}$ deduced from the Bombieri–Vinogradov theorem over ideals [12, (4.3)] through $d \le \mathrm{N}(\mathfrak{a}) \le d^2$. Theorem 2.3 (with Remark 2.4, since $\mathfrak{a} \mid \Phi_\ell(\pi)$ is a union of at most $(\ell-1)^{\omega(\mathfrak{a})}$ residue classes modulo $\mathfrak{a}$) gives [12, (4.6)] at level $x^{1/2}L^{-B'}$ instead. The weighted sieve of [12, (4.7)–(4.9)] then yields, for $\gg x/(\log x)^2$ primes $p \le x$, $p \equiv 1 \pmod 4$,

$$\Omega(a) \le \Big\lfloor \frac{1}{\lambda} + \frac{\ell-1}{\delta} \Big\rfloor \qquad \text{whenever } 0 < \theta < \delta < \alpha \text{ and } G(\theta,\delta,\lambda) > 0,$$

where

$$G(\theta,\delta,\lambda) = f\Big(\frac{\alpha}{\theta}\Big) - \lambda \int_\theta^\delta F\Big(\frac{\alpha-t}{\theta}\Big)\Big(1-\frac{t}{\delta}\Big)\frac{\mathrm{d}t}{t},$$

now with $\alpha = 1/2 - \varepsilon$, where f, F are the lower and upper bound functions of the linear sieve, $F(s) = 2e^\gamma/s$ for $0 < s \le 3$ and $f(s) = 2e^\gamma \log(s-1)/s$ for $2 \le s \le 4$.[1] We choose $\alpha/4 \le \theta < \delta < \alpha$, so that $3 \le \alpha/\theta \le 4$ and $0 < (\alpha - t)/\theta \le 3$ on the whole range of integration; then only the closed forms are needed, and partial fractions give

$$G(\theta,\delta,\lambda) = \frac{2e^\gamma\theta}{\alpha}\Big[\log\Big(\frac{\alpha}{\theta}-1\Big) - \lambda\Big(\log\frac{\delta}{\theta} - \Big(\frac{\alpha}{\delta}-1\Big)\log\frac{\alpha-\theta}{\alpha-\delta}\Big)\Big].$$

At $\alpha = 0.499$, $\theta = 1/8$ this gives (interval arithmetic, but the reader may check it with a pocket calculator)

$$G\big(\tfrac18, \tfrac{9}{20}, \tfrac{10}{17}\big) = 0.4217\ldots > 0, \qquad G\big(\tfrac18, \tfrac{20}{41}, \tfrac{5}{6}\big) = 0.0253\ldots > 0.$$

The first choice gives $1/\lambda + 1/\delta = 353/90 < 4$, the second $1/\lambda + (\ell-1)/\delta = 2.05(\ell-1) + 1.2$ (when this is an integer, as for $\ell = 17$, increase λ slightly, which keeps $G > 0$ and makes the bound strict). Since the elements $\mathrm{N}(\pi+1)/4 < x$ have no prime factor below x^θ and are therefore square-free after the removal step of [12, p. 27], we obtain:

Corollary 3.3. *Let E be the curve $y^2 = x^3 - x$. For sufficiently large x,*

$$\#\Big\{p \le x :\ p \equiv 1 \pmod 4,\ \frac{|E(\mathbb{F}_{p^2})|}{4|E(\mathbb{F}_p)|} = P_3\Big\} \gg \frac{x}{(\log x)^2},$$

and for every odd prime $\ell \ge 3$,

$$\#\Big\{p \le x :\ p \equiv 1 \pmod 4,\ \Omega\Big(\frac{|E(\mathbb{F}_{p^\ell})|}{|E(\mathbb{F}_p)|}\Big) \le \lfloor 2.05(\ell-1) + 1.2 \rfloor\Big\} \gg \frac{x}{(\log x)^2}.$$

This improves P_5 and $\lfloor 4.1(\ell-1)+1.2\rfloor$ in [12, Theorem 1.3]; for $\ell = 3, 5, 7$ the bounds are $5, 9, 13$ in place of $9, 17, 25$. (For $|E(\mathbb{F}_p)|/8$ itself the same computation gives P_3, which is superseded by (1).) Figure 1 shows how the quantity $\min(1/\lambda + w/\delta)$, minimised over θ, δ with λ chosen so that $G = 0$, decreases with the level α; the gain from $\alpha = 1/4$ to $\alpha = 1/2$ is a factor close to 2 for large ℓ, as the term $(\ell-1)/\delta$ with $\delta \to \alpha$ dominates. The vector-sieve results [12, Theorems 1.4, 1.5] also depend on the level through [12, Proposition 2.8]; we have not recomputed them.

[1] In [12, (4.11)] the value $F(s) = 2e^\gamma$ is used for $0 < s < 1$. The upper bound sieve at level $D < z$ is obtained by sifting only by the primes below D, which gives $F(s) = 2e^\gamma/s$ also for $0 < s < 1$; the value $2e^\gamma$ is not an admissible upper bound there. With $F(s) = 2e^\gamma/s$ the quantity $G(1/15, 1/4.1, 1/1.3)$ at level $1/4.01$ in [12] equals 0.0676 rather than 0.1341; it is still positive, so the conclusions of [12, §4] are unaffected.

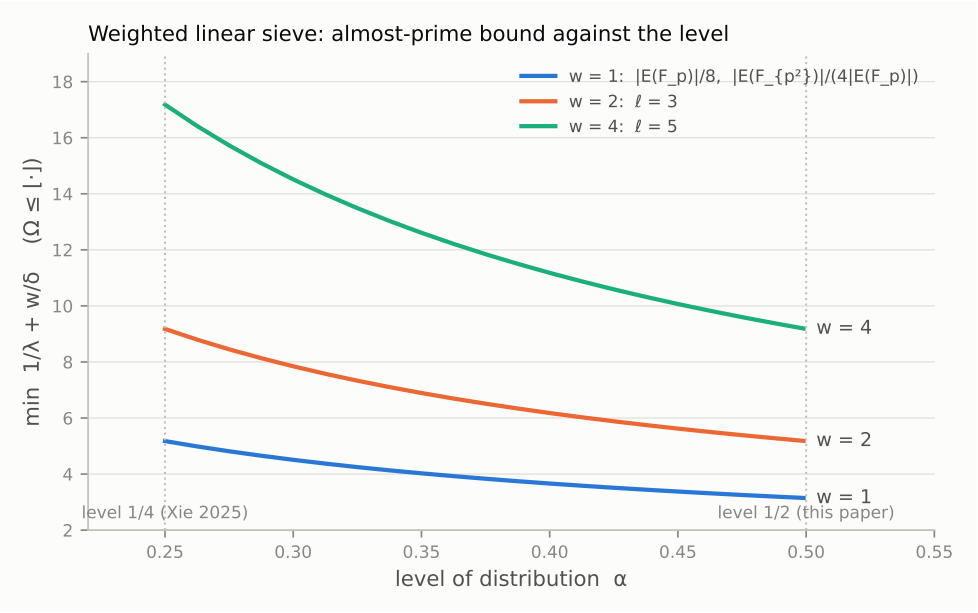


FIGURE 1. The almost-prime bound $\min_{\theta,\delta}(1/\lambda + w/\delta)$ of the weighted linear sieve [12, (4.7)–(4.9)] as a function of the level of distribution α, with λ chosen so that $G(\theta,\delta,\lambda) = 0$ and F, f continued beyond their closed-form ranges by the delay-differential equations $(sF(s))' = f(s-1)$, $(sf(s))' = F(s-1)$. The weight w is 1 for $|E(\mathbb{F}_p)|/8$ and $|E(\mathbb{F}_{p^2})|/(4|E(\mathbb{F}_p)|)$, and $\ell-1$ for $|E(\mathbb{F}_{p^\ell})|/|E(\mathbb{F}_p)|$. The bound on Ω is the integer part. Floating-point exploration (`scan_levels.py` in the repository described at the end of the paper); the values used in Corollary 3.3 are certified separately.

4. Remarks on the remaining parts of the proof in [7]

Since [12, Remark 4.1] casts doubt on (1), we record the points of [7] that we have checked besides the level of distribution.

4.1. **Proposition 5.1 of [7].** This is the Bombieri–Vinogradov theorem for $\Omega(x)$, i.e. hypothesis (i) for $R_{\mathcal{B}}$. Its proof in [7] is a standard dispersion argument: after separating the variable π_3 from $\pi_1\pi_2$ by a partition of $[z,y)$ into intervals $[z_k, z_k(1+\delta))$, $\delta = L^{-A-2}$, the character sums factor as $A_k(\chi)B_k(\chi)$ with $A_k(\chi) = \sum_{\mathrm{N}(\pi_3)\in[z_k,z_k(1+\delta))}\chi(\pi_3)$; after reduction to primitive characters, conductors of norm $> R = L^{O_A(1)}$ are treated by the large sieve for $\mathbb{Z}[i]$ [7, Lemma 5.5] (Huxley [5]) and Cauchy's inequality, and conductors of norm $\le R$ by [7, Lemma 5.2]. We have re-derived the proposition in full; two comments are in order.

(a) The hypothesis (5.3) of [7, Lemma 5.2] is attributed there to [7, Proposition 3.2], which is an average over moduli and does not give a bound for an individual modulus. What is needed is a Siegel–Walfisz theorem for primary Gaussian primes: for every $C > 0$, every odd ideal $\mathfrak{f} \neq (1)$ with $\mathrm{N}(\mathfrak{f}) \le (\log t)^C$ and every primitive character χ modulo $\mathfrak{f}$,

$$\sum_{\substack{\pi \text{ primary} \\ \mathrm{N}(\pi)\le t}} \chi(\pi) \ll_C t\exp(-c\sqrt{\log t}). \tag{6}$$

This holds (with an ineffective constant, as usual). Indeed, reduction modulo $2+2i$ maps the unit group $\{\pm1,\pm i\}$ isomorphically onto $(\mathbb{Z}[i]/(2+2i))^\times$, so there is exactly one character ψ

modulo $2+2i$ with $\psi(u) = \chi(u)^{-1}$ for all units u. The character $\chi\psi$ of $(\mathbb{Z}[i]/\mathfrak{f}(2+2i))^\times$ is trivial on the units, hence is a ray class character modulo $\mathfrak{f}(2+2i)$, i.e. a function on ideals; and for an odd prime ideal $\mathfrak{p}$ with primary generator π we have $(\chi\psi)(\mathfrak{p}) = \chi(\pi)\psi(\pi) = \chi(\pi)$ because $\pi \equiv 1 \pmod{2+2i}$. Thus the left side of (6) is exactly $\sum_{\mathrm{N}(\mathfrak{p})\le t}(\chi\psi)(\mathfrak{p})$ (up to the prime $(1+i)$), and $\chi\psi$ is non-principal: its restriction to $(\mathbb{Z}[i]/\mathfrak{f})^\times \times \{1\}$ is $\chi \neq \chi_0$. Now (6) is Lemma 16 of Mitsui [11] (equivalently, the Theorem at the end of [11, §1]): for a fixed number field K, a constant $C > 0$, and a character χ of the ray class group modulo $\mathfrak{q}$ with $\mathrm{N}(\mathfrak{q}) \le (\log t)^C$,

$$\sum_{\mathrm{N}(\mathfrak{p})\le t} \chi(\mathfrak{p}) = E(\chi)\int_2^t \frac{\mathrm{d}u}{\log u} + O\big(t\exp(-c\sqrt{\log t})\big),$$

where $E(\chi) = 1$ if χ is principal and 0 otherwise, and $c > 0$ and the implied constant depend only on K and C; the Siegel-type zero-free region for real characters that this requires is [11, Lemma 11] (see also Fogels [2] and Hinz [4]). In the proof of [7, Proposition 5.1] the hypothesis (5.3) is only used for moduli of norm $\le R\cdot\Delta^{-6} = L^{O_A(1)}$, so (6) suffices.

(b) For the prime variable π_3 the coprimality condition $(\pi_3, \mathfrak{a}) = 1$ in [7, (5.5)] is trivial: $\mathrm{N}(\pi_3) \ge z$ exceeds the norm of any prime factor of a modulus of norm $\le L^{O(1)}$, so $(\pi_3,\mathfrak{a}) \neq 1$ means $\pi_3 \mid \mathfrak{a}$, which happens for at most $\omega(\mathfrak{a}) \ll L$ primes. Hence [7, Lemma 5.2] can be replaced by (6) directly, and the bound for the small conductors is $\ll xRL^2\cdot L^{-C'}$ for any C'. For the large conductors one obtains, exactly as in [7, (5.8)], the bound $x^{1/2}L^{O(1)}\big(Q + y^{1/2} + (x/z)^{1/2} + x^{1/2}/R\big)$, which is $\ll xL^{-A}$ for $Q = x^{1/2}L^{-B}$ and suitable B, R. The exponents $1/8$ and $1/3$ enter only through $y^{1/2}x^{1/2} = x^{2/3}$ and $(x/z)^{1/2}x^{1/2} = x^{15/16}$.

4.2. **Minor points.** In the derivation of [7, (5.9)] the condition $(\omega,\mathfrak{a}) = 1$ is removed from the main term; this costs $\Phi(\mathfrak{a})^{-1}\#\{\omega : (\omega,\mathfrak{a}) \neq 1\}$ per modulus, which is why the weight $\Phi(\mathfrak{a})^{-1}$ must be kept in the sum over $\mathfrak{a}$ (as above); without it the sum would be of size xQ/z. The elements of $\mathcal{B}(x)$ are odd (not "divisible by 2" as stated on [7, p. 830]), which is what is used: the sieve for W_3 runs over odd primes $p < \sqrt{x}$, and $\prod_{2<p<\sqrt{x}}(1-g(p)) = (1-g(2))^{-1}\cdot\frac{\log z}{\log\sqrt{x}}\,V(z)(1+o(1)) = \frac12 V(z)(1+o(1))$ by [7, (3.10)] and $g(2) = \frac12$. The constants $\frac12 e^\gamma\log 6$ in [7, (4.3)] (the sum over p_0 runs over split primes only, which have density $\frac12$ among all primes, with $g(p_0)\sim 2/p_0$) and $c < 0.36308373$ in [7, (5.10)] were recomputed. The Chen-type weight [7, (2.2)] is the classical one: an element a of $\mathcal{A}(x)$ coprime to $2P(z)Q(z)$ with positive weight is either P_2 or not square-free, because $a \le x/8$ has at most one prime factor $\le y$ unless two such factors give weight ≤ 0, and a square-free a with exactly one prime factor $\le y$ and at least three prime factors is of the form $p_1p_2p_3$ counted by W_3 (four factors would exceed $x^{1/8}\cdot x = x^{9/8}$); the non-square-free elements are $\ll x\log x/(z\log z)$ in number.

Data and code availability

The scripts behind the numerical statements of this note are available at

https://github.com/Ruqing1963/cm-orders-level-half

and archived at Zenodo, doi:10.5281/zenodo.23428952. They comprise the interval-arithmetic evaluation of G in Section 3 (`certify_G.py`, exact rational parameters, closed form), the computation of the constants of Section 3.4, the exploration of Figure 1 (`scan_levels.py`, with the resulting table), and a first-principles check of the bookkeeping of Section 3. The latter script, `check_omega_distribution.py`, builds for $x = 10^8$ the set $\Omega(x)$ and the primary primes of norm $\le x$, verifies the identity (3) exactly for $p = 5, 13, 17, 29, 37, 41$, compares the

divisor densities with $g(d)$, and measures the discrepancies $E(x;\alpha,\mathfrak{a})$, which are of the size of the square root of the main term for all moduli tested.

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